



Indiana State Standards Alignment

For Grades K-8

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About Zearn

Zearn is the 501(c)(3) nonprofit educational organization behind Zearn Math, the top-rated math learning platform used by 1 in 4 elementary-school students and by more than 1 million middle-school students nationwide. Zearn Math supports teachers with research-backed curriculum and digital lessons proven to double student academic growth across all levels of student proficiency. Zearn Math can be used flexibly to support a range of instructional needs including Tier 1 and 2 support, acceleration, intervention, tutoring, after school programming and summer programming.

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Zearn Math is aligned to Indiana's Academic Standards for Mathematics.

We are committed to ensuring that our high-quality instructional materials align to Indiana's Academic Standards for Mathematics to fully support Indiana's students and teachers. The tables on the following pages, organized by Indiana Academic Standards, identify the alignment between each grade-level standard and Zearn Math lesson(s).

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Kindergarten

INDIANA'S ACADEMIC STANDARDS FOR MATHEMATICS		ZEARN MATH
Kindergarten Standards		Lessons
Number Sense		
K.NS.1	Count to at least 100 by ones and tens. Count by one from any given number.	Mission 5, Topic C, Lesson 13 Mission 5, Topic D, Lesson 15-18
K.NS.2	Write whole numbers from 0 to 20 and identify number words from 0 to 10. Represent a number of objects with a written numeral 0-20 (with 0 representing a count of no objects).	Mission 1, Topic D, Lesson 12-16 Mission 1, Topic E, Lesson 17-22 Mission 1, Topic F, Lesson 24, 26-28 Mission 1, Topic G, Lesson 29-32 Mission 1, Topic H, Lesson 33-35, 37 Mission 4, Topic A, Lesson 1-6 Mission 4, Topic B, Lesson 7-12 Mission 4, Topic E, Lesson 25, 27 Mission 5, Topic A, Lesson 1-4 Mission 5, Topic B, Lesson 6, 9 Mission 5, Topic C, Lesson 10-13
K.NS.3	Say the number names in standard order when counting objects, pairing each object with one and only one number name and each number name with one and only one object. Understand that the last number name said describes the number of objects counted and that the number of objects is the same regardless of their arrangement or the order in which they were counted. Count out the number of objects, given a number from 1 to 20.	Mission 1, Topic B, Lesson 5-6 Mission 1, Topic C, Lesson 7-11 Mission 1, Topic D, Lesson 12-16 Mission 1, Topic E, Lesson 17-22 Mission 1, Topic F, Lesson 23-28 Mission 1, Topic G, Lesson 29-32 Mission 1, Topic H, Lesson 33-37 Mission 5, Topic A, Lesson 1-5 Mission 5, Topic B, Lesson 6
K.NS.4	Identify sets of 1 to 10 objects in patterned arrangements and tell how many without counting.	<i>Digital Activities</i> <i>Zearn Math includes digital activities that allow for instantly recognizing a quantity in a structured arrangement.</i>
K.NS.5	Identify whether the number of objects in one group is greater than, less than, or equal to the number of objects in another group (e.g., by using matching and counting strategies).	Mission 3, Topic E, Lesson 17-19 Mission 3, Topic F, Lesson 20-24 Mission 3, Topic G, Lesson 25-28

INDIANA'S ACADEMIC STANDARDS FOR MATHEMATICS		ZEARN MATH
Kindergarten Standards		Lessons
K.NS.6	Compare the values of two numbers from 1 to 20 presented as written numerals.	Mission 3, Topic F, Lesson 20 Mission 3, Topic G, Lesson 25-28 <i>Zearn Math lessons focus on comparing numbers to 10, not 20. Teachers will need to amend instruction to ensure students master the full intent of this standard.</i>
K.NS.7	Define and model a "ten" as a group of ten ones. Model equivalent forms of whole numbers from 10 to 20 as groups of tens and ones using objects and drawings.	Mission 5, Topic A, Lesson 2-5 Mission 5, Topic B, Lesson 6-9 Mission 5, Topic C, Lesson 10-13 Mission 5, Topic E, Lesson 20-21, 23
Computation & Algebraic Thinking		
K.CA.1	Solve real-world problems that involve addition and subtraction within 10 using modeling with objects or drawings.	Mission 4, Topic C, Lesson 16-18 Mission 4, Topic D, Lesson 19, 21 Mission 4, Topic F, Lesson 31 Mission 4, Topic G, Lesson 34-36 Mission 4, Topic H, Lesson 37
K.CA.2	Use objects or drawings to model the decomposition of numbers less than 10 into pairs in more than one way. Identify corresponding equations.	Mission 1, Topic C, Lesson 9-11 Mission 1, Topic D, Lesson 14-15 Mission 4, Topic A, Lesson 2, 4-6 Mission 4, Topic B, Lesson 7-12 Mission 4, Topic D, Lesson 22-24 Mission 4, Topic E, Lesson 25-28 Mission 4, Topic F, Lesson 32
K.CA.3	Find the number that makes 10 when added to the given number for any number from 1 to 9 (e.g., by using objects or drawings), and record the answer with a drawing or an equation.	Mission 4, Topic H, Lesson 39-40
K.CA.4	Create, extend, and give an appropriate rule for simple repeating and growing patterns with numbers and shapes.	Mission 5, Topic C, Lesson 11-13
Geometry		

INDIANA'S ACADEMIC STANDARDS FOR MATHEMATICS		ZEARN MATH
Kindergarten Standards		Lessons
K.G.1	Compare two- and three-dimensional shapes in different sizes and orientations, using informal language to describe their similarities, differences, parts (e.g., number of sides and vertices/"corners"), and other attributes (e.g., having sides of equal length).	Mission 2, Topic A, Lesson 1-4 Mission 2, Topic B, Lesson 6-7 Mission 2, Topic C, Lesson 9-10 Mission 6, Topic A, Lesson 3
Measurement		
K.M.1	Make direct comparisons of the length, capacity, weight, and temperature of objects, and identify which object is shorter, longer, taller, lighter, heavier, warmer, cooler, or holds more.	Mission 3, Topic A, Lesson 1-3 Mission 3, Topic B, Lesson 4-7 Mission 3, Topic C, Lesson 8-12 Mission 3, Topic D, Lesson 13-15 Mission 3, Topic F, Lesson 20 Mission 3, Topic G, Lesson 28 Mission 3, Topic H, Lesson 29-32
K.M.2	Identify and use appropriate terms to describe intervals of time including: morning, afternoon, evening, today, yesterday, tomorrow, day, week, month, and year; describe how calendars and clocks are tools to measure time.	<i>It is recommended that teachers use supplemental materials to fully address this standard.</i>
Data Analysis		
K.DA.1	With guidance, collect and organize data into simple bar graphs, pictographs, and/or tables to identify patterns and make comparisons.	Mission 1, Topic A, Lesson 1-3 Mission 1, Topic B, Lesson 4-6 Mission 1, Topic C, Lesson 7 Mission 2, Topic A, Lesson 1 Mission 2, Topic B, Lesson 7 <i>It is recommended that teachers use supplemental materials to fully address this standard.</i>

1st Grade

INDIANA'S ACADEMIC STANDARDS FOR MATHEMATICS		ZEARN MATH
1st Grade Standards		Lessons
Number Sense		
1.NS.1	Count to at least 120 by ones, fives, and tens from any given number. In this range, read and write numerals and represent a number of objects with a written numeral.	Mission 4, Topic A, Lesson 1-6 Mission 6, Topic B, Lesson 4,7,9
1.NS.2	Model place value concepts of two-digit numbers, multiples of 10, and equivalent forms of whole numbers using objects and drawings	Mission 2, Topic D, Lesson 26-29 Mission 4, Topic A, Lesson 1-4, 6 Mission 4, Topic F, Lesson 23 Mission 6, Topic B, Lesson 3-4, 8-9
1.NS.3	Match the ordinal numbers (e.g., first, second, third) with an ordered set of up to 20 items.	<i>It is recommended that teachers use supplemental materials to fully address this standard.</i>
1.NS.4	Use place value understanding to compare two two-digit numbers based on meanings of the tens and ones digits, recording the results of comparisons with the symbols $>$, $=$, and $<$.	Mission 4, Topic B, Lesson 7-10 Mission 6, Topic B, Lesson 6
Computation & Algebraic Thinking		
1.CA.1	Demonstrate fluency with addition facts and the corresponding subtraction facts within 20. Use strategies such as counting on; making ten (e.g., $8 + 6 = 8 + 2 + 4 = 10 + 4 = 14$); decomposing a number leading to a 10 (e.g., $13 - 4 = 13 - 3 - 1 = 10 - 1 = 9$); using the relationship between addition and subtraction (e.g., knowing that $8 + 4 = 12$, one knows $12 - 8 = 4$); and creating equivalent but easier or known sums (e.g., adding $6 + 7$ by creating the known equivalent $6 + 6 + 1 = 12 + 1 = 13$). Model the role of 0 and the equal sign in addition and subtraction using objects or drawings.	Mission 1, Topic C, Lesson 9-13 Mission 1, Topic D, Lesson 14-16 Mission 1, Topic E, Lesson 17-20 Mission 1, Topic F, Lesson 21-24 Mission 1, Topic G, Lesson 26-27 Mission 1, Topic I, Lesson 33-37 Mission 1, Topic J, Lesson 38-39 Mission 2, Topic A, Lesson 1-11 Mission 2, Topic B, Lesson 12-21 Mission 2, Topic C, Lesson 25 Mission 2, Topic D, Lesson 27-29
1.CA.2	Solve real-world problems involving addition and subtraction within 20 in situations of adding to, taking from, putting together, taking apart, and comparing,	Mission 1, Topic B, Lesson 4-8 Mission 1, Topic C, Lesson 9-13 Mission 1, Topic G, Lesson 25

INDIANA'S ACADEMIC STANDARDS FOR MATHEMATICS		ZEARN MATH
1st Grade Standards		Lessons
	with unknowns in all parts of the addition or subtraction problem (e.g., by using objects, drawings, and equations with a symbol for the unknown number to represent the problem).	Mission 1, Topic H, Lesson 28-32 Mission 2, Topic A, Lesson 3-5, 7-9, 11 Mission 2, Topic B, Lesson 12-13, 17-18, 21 Mission 2, Topic C, Lesson 22-24 Mission 2, Topic D, Lesson 27-29 Mission 3, Topic D, Lesson 10-13 Mission 4, Topic E, Lesson 19-22 Mission 6, Topic A, Lesson 1-2 Mission 6, Topic F, Lesson 25-27
1.CA.3	Using number sense and place value strategies, add within 100, including adding a two-digit number and a one-digit number, and adding a two-digit number and a multiple of 10. Use models or drawings and strategies based on place value, properties of operations, and/or the relationship between addition and subtraction; describe the strategy and explain the reasoning used.	Mission 4, Topic C, Lesson 11-12 Mission 4, Topic D, Lesson 13-18 Mission 4, Topic F, Lesson 23-29 Mission 6, Topic C, Lesson 10-17 Mission 6, Topic D, Lesson 18-19
1.CA.4	Create, extend, and give an appropriate rule for number patterns using addition within 100.	Mission 1, Topic B, Lesson 5-6, 8 Mission 1, Topic D, Lesson 16 Mission 1, Topic F, Lesson 22-23 Mission 2, Topic A, Lesson 10 Mission 4, Topic D, Lesson 15 Mission 6, Topic B, Lesson 7
Geometry		
1.G.1	Distinguish between defining attributes of two- and three-dimensional shapes (e.g., triangles are closed and three-sided) versus non-defining attributes (e.g., color, orientation, overall size). Create and draw two-dimensional shapes with defining attributes.	Mission 5, Topic A, Lesson 1-3
1.G.2	Use two-dimensional shapes (e.g., rectangles, squares, trapezoids, triangles, half-circles, quarter-circles) or three-dimensional shapes (e.g., cubes, right rectangular prisms, right circular cones, and right circular cylinders) to create a composite shape, and compose new shapes from the composite shape. [In grade 1, students do not need to learn formal names such as "right rectangular prism."]	Mission 5, Topic B, Lesson 4-6

INDIANA'S ACADEMIC STANDARDS FOR MATHEMATICS		ZEARN MATH
1st Grade Standards		Lessons
1.G.3	Partition circles and rectangles into two and four equal parts; describe the parts using the words halves, fourths, and quarters; and use the phrases half of, fourth of, and quarter of. Describe the whole as two of, or four of, the parts. Understand for partitioning circles and rectangles into two and four equal parts that decomposing into equal parts creates smaller parts.	Mission 5, Topic C, Lesson 7-9 Mission 5, Topic D, Lesson 10
Measurement		
1.M.1	Use direct comparison or a nonstandard unit to compare and order objects according to length, area, capacity, weight, and temperature.	Mission 3, Topic A, Lesson 1-3 Mission 3, Topic B, Lesson 4-6 Mission 3, Topic C, Lesson 7-9
1.M.2	Tell and write time to the nearest half-hour and relate time to events (before/after, shorter/longer) using analog clocks. Explain how to read hours and minutes using digital clocks.	Mission 5, Topic D, Lesson 10-13
1.M.3	Identify the value of a penny, nickel, dime, and a collection of pennies, nickels, and dimes.	Grade K, Mission 5, Topic E, Lesson 20, 23 Grade K, Mission 6, Topic A, Lesson 1 Mission 4, Topic A, Lesson 6 Mission 6, Topic E, Lesson 20- 21
Data Analysis		
1.DA.1	With guidance, collect data from a simple survey or collaborative investigation; organize data into appropriate single-unit bar graphs, pictographs, and/or tables and draw conclusions based on mathematical observations, comparisons, and grade-level computation strategies.	<i>It is recommended that teachers use supplemental materials to fully address this standard.</i>

2nd Grade

INDIANA'S ACADEMIC STANDARDS FOR MATHEMATICS		ZEARN MATH
2nd Grade Standards		Lessons
Number Sense		
2.NS.1	Count by ones, twos, fives, tens, and hundreds up to at least 1,000 from any given number.	Mission 3, Topic B, Lesson 2-3 Mission 3, Topic C, Lesson 4 Mission 3, Topic D, Lesson 8-9 Mission 3, Topic E, Lesson 12-15 Mission 3, Topic G, Lesson 19-21 Mission 7, Topic E, Lesson 21 Mission 8, Topic D, Lesson 14
2.NS.2	Read and write whole numbers up to 1,000. Use words, models, standard form, and expanded form to represent and show equivalent forms of whole numbers up to 1,000.	Mission 3, Topic C, Lesson 4-7 Mission 3, Topic D, Lesson 8-9 Mission 3, Topic E, Lesson 13-14
2.NS.3	Determine whether a group of objects (up to 20) has an odd or even number of members (e.g., by placing that number of objects in two groups of the same size and recognizing that for even numbers no object will be left over and for odd numbers one object will be left over, or by pairing objects or counting them by twos).	Mission 6, Topic D, Lesson 17-19
2.NS.4	Define and model a "hundred" as a group of ten tens. Model place value concepts of three-digit numbers, multiples of 100, and equivalent forms of whole numbers using objects and drawings.	Mission 3, Topic A, Lesson 1 Mission 3, Topic B, Lesson 2-3 Mission 3, Topic C, Lesson 4-7 Mission 3, Topic D, Lesson 8-9 Mission 3, Topic E, Lesson 11-15
2.NS.5	Use place value understanding to compare two three-digit numbers based on meanings of the hundreds, tens, and ones digits, using $>$, $=$, and $<$ symbols to record the results of comparisons.	Mission 3, Topic F, Lesson 16-18
Computation & Algebraic Thinking		
2.CA.1	Solve real-world problems involving addition and subtraction within 100 in situations of adding to, taking from, putting together, taking apart, and	Mission 1, Topic B, Lesson 3, 5-6, 8 Mission 4, Topic A, Lesson 1-5 Mission 4, Topic B, Lesson 6-10

INDIANA'S ACADEMIC STANDARDS FOR MATHEMATICS		ZEARN MATH
2nd Grade Standards		Lessons
	comparing, with unknowns in all parts of the addition or subtraction problem (e.g., by using drawings and equations with a symbol for the unknown number to represent the problem). Use estimation to decide whether answers are reasonable in addition problems.	Mission 4, Topic C, Lesson 11-14, 16 Mission 4, Topic F, Lesson 31 Mission 7, Topic B, Lesson 7-8, 12-13
2.CA.2	Using number sense and place value strategies, add and subtract within 1,000, including composing and decomposing tens and hundreds. Use models, drawings, and strategies based on place value, properties of operations, and/or the relationship between addition and subtraction; describe the strategy and explain the reasoning used.	Mission 4, Topic B, Lesson 9 Mission 4, Topic C, Lesson 14-15 Mission 4, Topic D, Lesson 17-22 Mission 4, Topic E, Lesson 23-28 Mission 4, Topic F, Lesson 29-30 Mission 5, Topic A, Lesson 1-7 Mission 5, Topic B, Lesson 8-12 Mission 5, Topic C, Lesson 13-18 Mission 5, Topic D, Lesson 19-20
2.CA.3	Show that the order in which two numbers are added (commutative property) and how the numbers are grouped in addition (associative property) will not change the sum. These properties can be used to show that numbers can be added in any order.	Grade 1, Mission 1, Topic E, Lesson 19-20 Grade 1, Mission 1, Topic F, Lesson 21-24 Grade 1, Mission 2, Topic A, Lesson 2, 6, 10-11
2.CA.4	Create, extend, and give an appropriate rule for number patterns using addition and subtraction within 1,000.	Mission 3, Topic G, Lesson 21 Mission 4, Topic A, Lesson 1 Mission 5, Topic A, Lesson 1 Mission 6, Topic C, Lesson 16
Geometry		
2.G.1	Identify, describe, and classify two- and three-dimensional shapes (i.e., triangle, square, rectangle, cube, right rectangular prism) according to the number and shape of faces and the number of sides and/or vertices. Draw two-dimensional shapes.	Mission 8, Topic A, Lesson 1-5 Mission 8, Topic B, Lesson 6
2.G.2	Investigate and predict the result of composing and decomposing two- and three-dimensional shapes.	<i>It is recommended that teachers use supplemental materials to fully address this standard.</i>
2.G.3	Partition a rectangle into rows and columns of same-size (unit) squares and count to find the total number of same-size squares.	Mission 6, Topic C, Lesson 14-15

INDIANA'S ACADEMIC STANDARDS FOR MATHEMATICS		ZEARN MATH
2nd Grade Standards		Lessons
2.G.4	Partition circles and rectangles into two, three, or four equal parts; describe the shares using the words halves, thirds, half of, a third of, etc.; and describe the whole as two halves, three thirds, or four fourths. Recognize that equal parts of identical wholes need not have the same shape.	Mission 8, Topic B, Lesson 7-8 Mission 8, Topic C, Lesson 9-12 Mission 8, Topic D, Lesson 13
Measurement		
2.M.1	Describe the relationships among an inch, foot, and yard. Describe the relationship between a centimeter and meter.	Mission 7, Topic C, Lesson 15 Mission 7, Topic D, Lesson 16-17
2.M.2	Estimate and measure the length of an object by selecting and using appropriate tools, such as rulers, yardsticks, meter sticks, and measuring tapes to the nearest inch, foot, yard, centimeter, and meter.	Mission 2, Topic A, Lesson 1-3 Mission 2, Topic B, Lesson 4-5 Mission 2, Topic C, Lesson 6 Mission 2, Topic D, Lesson 9 Mission 7, Topic C, Lesson 14-17
2.M.3	Estimate and measure volume (capacity) using cups and pints. Add and subtract to solve real-world problems involving capacities that are given in the same units or obtained through investigations.	<i>It is recommended that teachers use supplemental materials to fully address this standard.</i>
2.M.4	Tell and write time to the nearest five minutes from analog clocks, using a.m. and p.m. Solve real-world problems involving addition and subtraction of time intervals on the hour or half hour.	Mission 8, Topic D, Lesson 13-16
2.M.5	Describe relationships of time, including seconds in a minute; minutes in an hour; hours in a day; days in a week; and days, weeks, and months in a year.	<i>It is recommended that teachers use supplemental materials to fully address this standard.</i>
2.M.6	Find the value of a collection of pennies, nickels, dimes, quarters, and dollars.	Mission 3, Topic D, Lesson 10 Mission 7, Topic B, Lesson 6-10, 12-13
Data Analysis		
2.DA.1	Collect, organize, and graph data from observations, surveys, and investigations using scaled bar graphs and pictographs (limit scale to 2s, 5s, 10s, and 100s); interpret mathematical relationships within the data	Mission 7, Topic A, Lesson 1-5 <i>It is recommended that teachers use supplemental materials to fully address this standard.</i>

INDIANA'S ACADEMIC STANDARDS FOR MATHEMATICS	ZEARN MATH
2nd Grade Standards	Lessons
using grade-level addition, subtraction, and comparison strategies.	

3rd Grade

INDIANA'S ACADEMIC STANDARDS FOR MATHEMATICS		ZEARN MATH
3rd Grade Standards		Lessons
Number Sense		
3.NS.1	Read and write whole numbers up to 10,000. Use words, models, standard form, and expanded form to represent and show equivalent forms of whole numbers up to 10,000.	<i>Digital Activities</i> <i>Zearn Math includes digital activities that focus on writing numbers in standard form, expanded form, and word form.</i>
3.NS.2	Model unit fractions as the quantity formed by 1 part when a whole is partitioned into equal parts; model non-unit fractions as the quantity formed by iterations of unit fractions. [In grade 3, limit denominators of fractions to 2, 3, 4, 6, 8.]	Mission 5, Topic B, Lesson 5-9 Mission 5, Topic C, Lesson 10, 12-13 Mission 5, Topic E, Lesson 20, 22, 24-25, 27 Mission 5, Topic F, Lesson 28-29
3.NS.3	Model a non-unit fraction on a number line by marking equal lengths from 0, identifying each part as a unit fraction and locating the non-unit fraction as the endpoint on the number line.	Mission 5, Topic D, Lesson 14-17, 19 Mission 5, Topic E, Lesson 21-23, 26
3.NS.4	Use fraction models to represent two simple equivalent fractions with attention to how the number and size of the parts differ even though the quantities are the same. Use this principle to generate simple equivalent fractions (e.g., $1/2 = 2/4$, $4/6 = 2/3$).	Mission 5, Topic C, Lesson 10-11, 13 Mission 5, Topic D, Lesson 16-19 Mission 5, Topic E, Lesson 20-27 Mission 5, Topic F, Lesson 28-29
3.NS.5	Compare two fractions with the same numerator or the same denominator by reasoning about their size based on the same whole. Record the results of comparisons with the symbols $>$, $=$, or $<$, and justify the conclusions (e.g., by using a visual fraction model).	Mission 5, Topic C, Lesson 10-11, 13 Mission 5, Topic D, Lesson 18-19 Mission 5, Topic F, Lesson 28-29
3.NS.6	Use place value understanding to round two- and three-digit whole numbers to the nearest 10 or 100.	Mission 2, Topic C, Lesson 12-14 Mission 2, Topic D, Lesson 17 Mission 2, Topic E, Lesson 20-21
Computation & Algebraic Thinking		

INDIANA'S ACADEMIC STANDARDS FOR MATHEMATICS		ZEARN MATH
3rd Grade Standards		Lessons
3.CA.1	Fluently add and subtract multi-digit whole numbers using strategies and algorithms based on place value, properties of operations, and relationships between addition and subtraction.	Mission 2, Topic A, Lesson 4-5 Mission 2, Topic B, Lesson 10-11 Mission 2, Topic D, Lesson 15-17 Mission 2, Topic E, Lesson 18-21
3.CA.2	Solve real-world problems involving addition and subtraction of multi-digit whole numbers (e.g., by using drawings and equations with a symbol for the unknown number to represent the problem).	Mission 1, Topic F, Lesson 21 Mission 3, Topic E, Lesson 18 Mission 7, Topic A, Lesson 1-3
3.CA.3	Model the concept of multiplication of whole numbers using equal-sized groups, arrays, area models, and equal intervals on a number line. Model the properties of 0 and 1 in multiplication using objects or drawings.	Mission 1, Topic A, Lesson 1-3 Mission 1, Topic B, Lesson 6 Mission 1, Topic C, Lesson 7-10 Mission 1, Topic E, Lesson 15, 17 Mission 3, Topic D, Lesson 12 Mission 3, Topic E, Lesson 16
3.CA.4	Model the concept of division of whole numbers with the following models: partitioning, 6 sharing, and an inverse of multiplication. Model the properties of 0 and 1 in division using objects or drawings.	Mission 1, Topic D, Lesson 11-13 Mission 1, Topic E, Lesson 17 Mission 3, Topic E, Lesson 16-18
3.CA.5	Multiply and divide within 100 using strategies such as the relationship between multiplication and division (e.g., knowing that $8 \times 5 = 40$, one knows $40 \div 5 = 8$) or properties of operations.	Mission 1, Topic D, Lesson 11-13 Mission 1, Topic E, Lesson 14, 16 Mission 3, Topic B, Lesson 4-5 Mission 3, Topic C, Lesson 10 Mission 3, Topic D, Lesson 12-13 Mission 3, Topic E, Lesson 16-17 Mission 3, Topic F, Lesson 19
3.CA.6	Demonstrate fluency with mastery of multiplication facts and corresponding division facts of 0 to 10.	Mission 1, Topic D, Lesson 11-13 Mission 1, Topic E, Lesson 14, 16 Mission 3, Topic B, Lesson 4-5 Mission 3, Topic C, Lesson 10 Mission 3, Topic D, Lesson 12-13 Mission 3, Topic E, Lesson 16-17 Mission 3, Topic F, Lesson 19
3.CA.7	Solve real-world problems involving whole number multiplication and division within 100 in situations involving equal groups, arrays, and measurement quantities (e.g., by using drawings and equations with a symbol for the unknown number to represent the problem).	Mission 1, Topic B, Lesson 4-6 Mission 1, Topic C, Lesson 8-9 Mission 1, Topic D, Lesson 11-13 Mission 1, Topic E, Lesson 14-17 Mission 1, Topic F, Lesson 18 Mission 3, Topic A, Lesson 2-3

INDIANA'S ACADEMIC STANDARDS FOR MATHEMATICS		ZEARN MATH
3rd Grade Standards		Lessons
		Mission 3, Topic B, Lesson 7 Mission 3, Topic C, Lesson 11 Mission 3, Topic D, Lesson 15
3.CA.8	Create, extend, and give an appropriate rule for number patterns within 100 (including patterns in the addition table or multiplication table).	Mission 3, Topic A, Lesson 1 Mission 3, Topic D, Lesson 13-14 Mission 3, Topic E, Lesson 16-17 Mission 3, Topic F, Lesson 19
Geometry		
3.G.1	Define, identify, and classify four-sided shapes such as rhombuses, rectangles, and squares as quadrilaterals. Identify and draw examples and non-examples of quadrilaterals.	Mission 7, Topic B, Lesson 4-9
3.G.2	Identify, describe, and draw points, lines, and line segments using appropriate tools (e.g., ruler, straightedge, and technology), and use these terms when describing two-dimensional shapes.	Grade 4, Mission 4, Topic D, Lesson 13-15
3.G.3	Partition shapes into parts with equal areas. Express the area of each part as a unit fraction of the whole (i.e., $\frac{1}{2}$, $\frac{1}{3}$, $\frac{1}{4}$, $\frac{1}{6}$, $\frac{1}{8}$).	Mission 5, Topic A, Lesson 1-4 Mission 5, Topic B, Lesson 5-9 Mission 5, Topic C, Lesson 10-13 Mission 5, Topic E, Lesson 24-25, 27 Mission 5, Topic F, Lesson 28-29
Measurement		
3.M.1	Estimate and measure the mass of objects in grams (g) and kilograms (kg) and the volume of objects in quarts (qt), gallons (gal), and liters (l). Add, subtract, multiply, or divide to solve one-step, real-world problems involving masses or volumes that are given in the same units or obtained through investigation.	Mission 2, Topic B, Lesson 7-11 Mission 2, Topic C, Lesson 12 Mission 2, Topic D, Lesson 15-17 Mission 2, Topic E, Lesson 18-21 Mission 3, Topic B, Lesson 6
3.M.2	Choose and use appropriate units and tools to estimate and measure length, weight, and temperature. Estimate and measure length to a quarter-inch, weight in pounds, and temperature in degrees Celsius and Fahrenheit.	Mission 2, Topic B, Lesson 7-11 Mission 2, Topic C, Lesson 12 Mission 2, Topic D, Lesson 15-17 Mission 2, Topic E, Lesson 18-21 Mission 3, Topic B, Lesson 6 <i>It is recommended that teachers</i>

INDIANA'S ACADEMIC STANDARDS FOR MATHEMATICS		ZEARN MATH
3rd Grade Standards		Lessons
		<i>use supplemental materials to fully address this standard.</i>
3.M.3	Tell and write time to the nearest minute and measure time intervals in minutes. Solve word problems involving addition and subtraction of time intervals in minutes (e.g., by representing the problem on a number line diagram).	Mission 2, Topic A, Lesson 1-5 Mission 2, Topic D, Lesson 15-17 Mission 2, Topic E, Lesson 18, 21
3.M.4	Find the value of any collection of coins and bills. Write amounts less than a dollar using the ¢ symbol and write larger amounts using the \$ symbol in the form of dollars and cents (e.g., \$4.59). Solve real-world problems to determine whether there is enough money to make a purchase.	Grade 2, Mission 3, Topic D, Lesson 10 Grade 2, Mission 7, Topic B, Lesson 6-10, 12-13
3.M.5	Find the area of a rectangle with whole-number side lengths by modeling with unit squares, and show that the area is the same as would be found by multiplying the side lengths. Identify and draw rectangles with the same perimeter and different areas or with the same area and different perimeters.	Mission 4, Topic A, Lesson 1-4 Mission 4, Topic B, Lesson 5-8 Mission 4, Topic C, Lesson 9-11 Mission 4, Topic D, Lesson 12-13 Mission 7, Topic D, Lesson 18-22
3.M.6	Find perimeters of polygons given the side lengths or given an unknown side length.	Mission 7, Topic C, Lesson 10-17 Mission 7, Topic E, Lesson 23-30
Data Analysis		
3.DA.1	Collect, organize, and graph data from observations, surveys, and experiments using scaled bar graphs and pictographs. Solve real-world problems by analyzing and interpreting the data using grade-level computation and comparison strategies.	Mission 6, Topic A, Lesson 1-4

4th Grade

INDIANA'S ACADEMIC STANDARDS FOR MATHEMATICS		ZEARN MATH
4th Grade Standards		Lessons
Number Sense		
4.NS.1	Read and write whole numbers up to 1,000,000. Use words, models, standard form, and expanded form to represent and show equivalent forms of whole numbers up to 1,000,000.	Mission 1, Topic A, Lesson 3 Mission 1, Topic B, Lesson 4-6
4.NS.2	Model mixed numbers and improper fractions using visual fraction models such as number lines and area models. Use a visual fraction model to show the equivalency between whole numbers and whole numbers as fractions.	Grade 3, Mission 5, Topic D, Lesson 16-17 Grade 3, Mission 5, Topic E, Lesson 21-27
4.NS.3	Use fraction models to represent two equivalent fractions with attention to how the number and size of the parts differ even though the fractions themselves are the same size. Use this principle to generate equivalent fractions. [In grade 4, limit denominators of fractions to 2, 3, 4, 5, 6, 8, 10, 25, 100.]	Mission 5, Topic B, Lesson 7-11 Mission 5, Topic C, Lesson 14-15 Mission 5, Topic D, Lesson 21 Mission 5, Topic E, Lesson 27 Mission 6, Topic B, Lesson 5 Mission 6, Topic D, Lesson 12 <i>Zearn Math includes lessons that target equivalent fractions, however the lessons do not cover the denominator of 25. Teachers will need to amend instruction to ensure students master the full intent of this standard.</i>
4.NS.4	Compare two fractions with different numerators and different denominators (e.g., by creating common denominators or numerators, or by comparing to a benchmark, such as 0, $\frac{1}{2}$, and 1). Explain why comparisons are valid only when the two fractions refer to the same whole. Record the results of comparisons with symbols $>$, $=$, or $<$, and justify the conclusions (e.g., by using a visual fraction model).	Mission 5, Topic C, Lesson 12-15 Mission 5, Topic E, Lesson 26-27
4.NS.5	Write tenths and hundredths in decimal and fraction notations. Use words, models, standard form, and expanded form to represent decimal numbers to hundredths. Mentally calculate fraction and decimal	Mission 6, Topic A, Lesson 1-3 Mission 6, Topic B, Lesson 4-6 Mission 6, Topic C, Lesson 11 Mission 6, Topic D, Lesson 12-14

INDIANA'S ACADEMIC STANDARDS FOR MATHEMATICS		ZEARN MATH
4th Grade Standards		Lessons
	equivalents for halves and fourths (e.g., $1/2 = 0.5 = 0.50$, $7/4 = 1\ 3/4 = 1.75$).	Mission 6, Topic E, Lesson 15
4.NS.6	Compare two decimals to hundredths by reasoning about their size based on the same whole. Record the results of comparisons with the symbols $>$, $=$, or $<$, and justify the conclusions (e.g., by using a visual model).	Mission 6, Topic C, Lesson 9-11
4.NS.7	Use place value understanding to round multi-digit whole numbers to any given place value.	Mission 1, Topic C, Lesson 7-10 Mission 1, Topic D, Lesson 12 Mission 1, Topic E, Lesson 16-17 Mission 1, Topic F, Lesson 19
Computation & Algebraic Thinking		
4.CA.1	Multiply a whole number of up to four digits by a one-digit whole number and multiply two two-digit numbers, using strategies based on place value and the properties of operations. Describe the strategy and explain the reasoning.	Mission 3, Topic B, Lesson 4-6 Mission 3, Topic C, Lesson 7-11 Mission 3, Topic H, Lesson 34-38
4.CA.2	Find whole-number quotients and remainders with up to four-digit dividends and one-digit divisors using strategies based on place value, the properties of operations, and/or the relationship between multiplication and division. Describe the strategy and explain the reasoning.	Mission 3, Topic E, Lesson 14-21 Mission 3, Topic G, Lesson 26-33
4.CA.3	Show how the order in which two numbers are multiplied (commutative property) and how numbers are grouped in multiplication (associative property) will not change the product. Use these properties to show that numbers can be multiplied in any order. Investigate and apply the distributive property.	Grade 3, Mission 1, Topic C, Lesson 7-10 Grade 3, Mission 1, Topic E, Lesson 15-16 Grade 3, Mission 1, Topic F, Lesson 18-19 Grade 3, Mission 3, Topic A, Lesson 1-2 Grade 3, Mission 3, Topic C, Lesson 9 Grade 3, Mission 3, Topic D, Lesson 12-14 Grade 3, Mission 3, Topic F, Lesson 20

INDIANA'S ACADEMIC STANDARDS FOR MATHEMATICS		ZEARN MATH
4th Grade Standards		Lessons
4.CA.4	Investigate the mathematical relationship between factors and multiples for whole numbers from 1-100, including the set of factors and multiples for given numbers. Identify sets of factors and multiples for any given whole number up to 100.	Mission 3, Topic F, Lesson 22-25
4.CA.5	Solve real-world problems with whole numbers involving multiplicative comparison (e.g., by using drawings and equations with a symbol for the unknown number to represent the problem), distinguishing multiplicative comparison from additive comparison. [In grade 4, division problems should not include a remainder.]	Mission 3, Topic A, Lesson 2-3 Mission 3, Topic B, Lesson 5-6 Mission 3, Topic C, Lesson 9-11 Mission 3, Topic D, Lesson 12-13 Mission 3, Topic G, Lesson 26 Mission 5, Topic G, Lesson 39-40 Mission 7, Topic A, Lesson 4 Mission 7, Topic C, Lesson 14
4.CA.6	Add and subtract fractions with common denominators using visual fraction models. Decompose non-unit fractions to represent them as iterations of unit fractions.	Mission 5, Topic A, Lesson 1-6 Mission 5, Topic D, Lesson 11-21 Mission 5, Topic E, Lesson 22-25 Mission 5, Topic F, Lesson 29-34
4.CA.7	Add and subtract mixed numbers with common denominators (e.g., by replacing each mixed number with an equivalent fraction and/or by using properties of operations and the relationship between addition and subtraction).	Mission 5, Topic F, Lesson 31-34
4.CA.8	Solve real-world problems involving addition and subtraction of fractions referring to the same whole and having common denominators (e.g., by using visual fraction models and equations to represent the problem).	Mission 5, Topic D, Lesson 19
4.CA.9	Describe the relationship between two terms and use it to find a second number when a first number is given. Generate a number pattern that follows a given rule.	Mission 5, Topic H, Lesson 41
Geometry		
4.G.1	Identify, describe, and draw parallelograms, rhombuses, and trapezoids using appropriate tools (e.g., ruler, straightedge, and technology).	Mission 4, Topic D, Lesson 15

INDIANA'S ACADEMIC STANDARDS FOR MATHEMATICS		ZEARN MATH
4th Grade Standards		Lessons
4.G.2	Identify, describe, and draw rays, angles (right, acute, obtuse), and perpendicular and parallel lines using appropriate tools (e.g., ruler, straightedge, and technology). Identify these in two-dimensional figures.	Mission 4, Topic A, Lesson 1-4 Mission 4, Topic D, Lesson 13-15
4.G.3	Classify triangles and quadrilaterals based on the presence or absence of parallel or perpendicular lines, or right, acute, or obtuse angles.	Mission 4, Topic D, Lesson 13-15
Measurement		
4.M.1	Measure length to the nearest quarter-inch, eighth-inch, and millimeter.	
4.M.2	Within given measurement systems, convert larger units to smaller units, including km, m, cm; kg, g; lb, oz; l, ml; hr, min, sec., and use these conversions to solve real-world problems.	Mission 2, Topic A, Lesson 1-3 Mission 2, Topic B, Lesson 4 Mission 4, Topic A, Lesson 4 Mission 7, Topic A, Lesson 1-3 Mission 7, Topic B, Lesson 6-9 Mission 7, Topic C, Lesson 12-13
4.M.3	Use the four operations to solve real-world problems involving distances, intervals of time, volumes, masses of objects, and money. Include addition and subtraction problems involving simple fractions and problems that require expressing measurements given in a larger unit in terms of a smaller unit.	Mission 2, Topic A, Lesson 1-3 Mission 2, Topic B, Lesson 5 Mission 5, Topic G, Lesson 40 Mission 6, Topic D, Lesson 14 Mission 6, Topic E, Lesson 15-16 Mission 7, Topic A, Lesson 1-5 Mission 7, Topic B, Lesson 6-11 Mission 7, Topic C, Lesson 14
4.M.4	Apply the area and perimeter formulas for rectangles to solve real-world and other mathematical problems. Investigate the area of complex shapes composed of rectangles by decomposing them into non-overlapping rectangles and adding the areas of the non-overlapping parts; apply this technique to solve real-world problems and other mathematical problems.	Mission 3, Topic A, Lesson 1-3 Mission 7, Topic D, Lesson 15-16 Grade 3, Mission 4, Topic D, Lesson 12-16
Data Analysis		

INDIANA'S ACADEMIC STANDARDS FOR MATHEMATICS		ZEARN MATH
4th Grade Standards		Lessons
4.DA.1	Formulate questions that can be addressed with data. Collect, organize, and graph data from observations, surveys, and experiments using line plots with whole number intervals, single- and scaled bar graphs, and frequency tables. Solve real-world problems by analyzing and interpreting the data using grade-level computation and comparison strategies.	<i>It is recommended that teachers use supplemental materials to fully address this standard.</i>
4.DA.2	Make a line plot to display a data set of measurements in fractions of a unit ($\frac{1}{2}$, $\frac{1}{4}$, $\frac{1}{8}$). Solve problems involving addition and subtraction of fractions by using data displayed in line plots.	Mission 5, Topic E, Lesson 28 Mission 5, Topic G, Lesson 40

5th Grade

INDIANA'S ACADEMIC STANDARDS FOR MATHEMATICS		ZEARN MATH
5th Grade Standards		Lessons
Number Sense		
5.NS.1	Use a number line to compare and order fractions, mixed numbers, and decimals to thousandths. Write the results using $>$, $=$, and $<$ symbols.	Mission 1, Topic B, Lesson 5-6 Mission 1, Topic D, Lesson 9-10 Mission 1, Topic E, Lesson 11-12 Mission 1, Topic F, Lesson 13-15
5.NS.2	Explain different interpretations of fractions, including as parts of a whole, parts of a set, and division of whole numbers by whole numbers.	Mission 4, Topic B, Lesson 2-5
5.NS.3	Explain patterns in the number of zeros of the product when multiplying a number by powers of 10, and explain patterns in the placement of the decimal point when a decimal is multiplied or divided by a power of 10. Use whole-number exponents to denote powers of 10.	Mission 1, Topic A, Lesson 1-4 Mission 2, Topic A, Lesson 1 Mission 2, Topic C, Lesson 10-12 Mission 2, Topic D, Lesson 13-14 Mission 2, Topic G, Lesson 24
5.NS.4	Model percents as parts of 100 using pictures or diagrams and identify the equivalent fraction.	Grade 6, Mission 3, Topic D, Lesson 10
Computation & Algebraic Thinking		
5.CA.1	Find whole-number quotients and remainders with up to four-digit dividends and two-digit divisors using strategies based on place value, the properties of operations, and/or the relationship between multiplication and division. Describe the strategy and explain the reasoning used.	Mission 2, Topic E, Lesson 16-18 Mission 2, Topic F, Lesson 19-23 Mission 2, Topic H, Lesson 28-29 Mission 4, Topic G, Lesson 30-31
5.CA.2	Solve real-world problems involving multiplication and division of whole numbers (e.g., by using equations to represent the problem). In division problems that involve a remainder, explain how the remainder affects the solution to the problem.	Grade 4, Mission 3, Topic D, Lesson 12-13 Grade 4, Mission 3, Topic E, Lesson 14, 19 Grade 4, Mission 3, Topic G, Lesson 31-32

INDIANA'S ACADEMIC STANDARDS FOR MATHEMATICS		ZEARN MATH
5th Grade Standards		Lessons
5.CA.3	Add and subtract fractions and mixed numbers with unlike denominators using strategies or the standard algorithm.	Mission 3, Topic A, Lesson 1-2 Mission 3, Topic B, Lesson 3-6 Mission 3, Topic C, Lesson 8-12 Mission 3, Topic D, Lesson 14
5.CA.4	Solve real-world problems involving addition and subtraction of fractions referring to the same whole, including cases of unlike denominators (e.g., by using visual fraction models and equations to represent the problem). Use benchmark fractions and number sense of fractions to estimate mentally and assess whether the answer is reasonable.	Mission 3, Topic B, Lesson 3-7 Mission 3, Topic C, Lesson 8-12 Mission 3, Topic D, Lesson 13-15 Mission 4, Topic D, Lesson 10-12
5.CA.5	Use visual fraction models to multiply a fraction by a fraction or a whole number.	Mission 4, Topic C, Lesson 6-9 Mission 4, Topic E, Lesson 13-15, 17-18 Mission 5, Topic C, Lesson 10-15
5.CA.6	Use visual fraction models and numbers to divide a fraction by a fraction or a whole number.	Mission 4, Topic G, Lesson 25-31, 33
5.CA.7	Solve real-world problems involving multiplication of fractions, including mixed numbers (e.g., by using visual fraction models and equations to represent the problem).	Mission 4, Topic C, Lesson 6-7 Mission 4, Topic D, Lesson 10-12 Mission 4, Topic E, Lesson 13-18 Mission 4, Topic F, Lesson 22-24 Mission 5, Topic C, Lesson 12-15
5.CA.8	Solve real-world problems involving division of fractions and mixed numbers (e.g., by using visual fraction models and equations to represent the problem).	Mission 4, Topic G, Lesson 25-28 Mission 4, Topic H, Lesson 33
5.CA.9	Add, subtract, multiply, and divide decimals to hundredths, using models or drawings and strategies based on place value or the properties of operations. Describe the strategy and explain the reasoning.	Mission 1, Topic D, Lesson 9-10 Mission 1, Topic E, Lesson 11-12 Mission 1, Topic F, Lesson 13-16 Mission 2, Topic C, Lesson 10-12 Mission 2, Topic D, Lesson 13-14 Mission 2, Topic G, Lesson 24-27 Mission 2, Topic H, Lesson 28-29 Mission 4, Topic E, Lesson 17-18 Mission 4, Topic G, Lesson 29-31

INDIANA'S ACADEMIC STANDARDS FOR MATHEMATICS		ZEARN MATH
5th Grade Standards		Lessons
5.CA.10	Solve real-world problems involving addition, subtraction, multiplication, and division with decimals to hundredths including problems that involve money in decimal notation (e.g., by using equations, models or drawings, and strategies based on place value or properties of operations to represent the problem).	Mission 1, Topic F, Lesson 16 Mission 2, Topic D, Lesson 15 Mission 4, Topic F, Lesson 24
5.CA.11	Represent real-world problems and equations by graphing ordered pairs in the first quadrant of the coordinate plane, and interpret coordinate values of points in the context of the situation.	Mission 6, Topic C, Lesson 13-17 Mission 6, Topic D, Lesson 18-20
Geometry		
5.G.1	Identify, describe, and draw triangles (right, acute, obtuse) and circles using appropriate tools (e.g., ruler or straightedge, compass, and technology). Define and model the relationship between radius and diameter	Grade 4, Mission 4, Topic D, Lesson 13-15
Measurement		
5.M.1	Convert among different-sized standard measurement units within a given measurement system and use these conversions in solving multi-step, real-world problems.	Mission 1, Topic A, Lesson 4 Mission 2, Topic D, Lesson 13-15 Mission 4, Topic C, Lesson 9 Mission 4, Topic E, Lesson 19-20 Mission 5, Topic B, Lesson 5
5.M.2	Find the area of a rectangle with fractional side lengths by modeling with unit squares of the appropriate unit fraction side lengths, and show that the area is the same as would be found by multiplying the side lengths. Multiply fractional side lengths to find areas of rectangles, and represent fraction products as rectangular areas.	Mission 4, Topic E, Lesson 13-15
5.M.3	Develop and use formulas for the area of triangles, parallelograms, and trapezoids. Solve real-world and other mathematical problems that involve perimeter and area of triangles, parallelograms, and trapezoids, using appropriate units for measures.	Grade 6, Mission 1, Topic A, Lesson 2-3 Grade 6, Mission 1, Topic B, Lesson 4-6 Grade 6, Mission 1, Topic C, Lesson 7-9

INDIANA'S ACADEMIC STANDARDS FOR MATHEMATICS		ZEARN MATH
5th Grade Standards		Lessons
5.M.4	Find the volume of a right rectangular prism with whole-number side lengths by packing it with unit cubes, and show that the volume is the same as would be found by multiplying the edge lengths or multiplying the height by the area of the base.	Mission 5, Topic B, Lesson 45387
5.M.5	Apply the formulas $V = l \times w \times h$ and $V = B \times h$ for right rectangular prisms to find volumes of right rectangular prisms with whole-number edge lengths to solve real-world problems and other mathematical problems.	Mission 5, Topic B, Lesson 6-9
Data Analysis		
5.DA.1	Formulate questions that can be addressed with categorical and numerical data and make predictions about the data. Collect, organize, and graph data from observations, surveys, and experiments using line plots with fractional intervals, histograms, or other graphical representations that appropriately represent the data set.	Mission 4, Topic A, Lesson 1 Mission 4, Topic D, Lesson 10 <i>It is recommended that teachers use supplemental materials to fully address this standard.</i>
5.DA.2	Calculate measures of central tendency (mean, median, and mode) to describe a data set. Analyze data sets to determine which measure of central tendency appropriately describes the distribution of data.	Grade 6, Mission 8, Topic C, Lesson 9-10 Grade 6, Mission 8, Topic D, Lesson 13-14

6th Grade

INDIANA'S ACADEMIC STANDARDS FOR MATHEMATICS		ZEARN MATH
6th Grade Standards		Lessons
Number Sense		
6.NS.1	Use positive and negative numbers to represent and compare quantities in real-world contexts, explaining the meaning of 0 in each situation.	Mission 7, Topic A, Lesson 1, 5
6.NS.2	Explain how opposite signs of numbers indicate locations on opposite sides of 0 on the number line; identify the opposite of the opposite of a number.	Mission 5, Topic D, Lesson 9-11
6.NS.3	Compare and order rational numbers and plot them on a number line. Write, interpret, and explain statements of order for rational numbers in real-world contexts.	Mission 7, Topic A, Lesson 2-4, 6-7 Mission 7, Topic B, Lesson 8 Mission 7, Topic C, Lesson 11-13. 15
6.NS.4	Solve real-world problems with positive fractions and decimals by using one or two operations.	Mission 5, Topic A, Lesson 1 Mission 5, Topic E, Lesson 14 Mission 6, Topic E, Lesson 16
6.NS.5	Apply the order of operations and properties of operations (i.e., identity, inverse, commutative properties of addition and multiplication, associative properties of addition and multiplication, and distributive property) to evaluate numerical expressions with nonnegative rational numbers, including those using grouping symbols, such as parentheses, and involving whole number exponents.	Mission 1, Topic F, Lesson 17-18 Mission 6, Topic C, Lesson 12-15
6.NS.6	Find the greatest common factor of two whole numbers less than or equal to 100 and the least common multiple of two whole numbers less than or equal to 12. Use the distributive property to express a sum of two whole numbers from 1 to 100, with a common factor as a multiple of a sum of two whole numbers with no common factor.	Mission 7, Topic D, Lesson 16-18
6.NS.7	Apply the properties of operations (i.e., identity, inverse, commutative, associative, distributive properties) to create equivalent linear expressions and	Mission 5, Topic D, Lesson 13 Mission 6, Topic B, Lesson 8, 10-11

INDIANA'S ACADEMIC STANDARDS FOR MATHEMATICS		ZEARN MATH
6th Grade Standards		Lessons
	to justify whether two linear expressions are equivalent when the two expressions name the same number regardless of which value is substituted into them.	
6.NS.8	Evaluate positive rational numbers with whole number exponents.	Mission 1, Topic F, Lesson 17-18 Mission 6, Topic C, Lesson 12-13
Ratios & Proportional Reasoning		
6.RP.1	Convert between any two representations (fractions, decimals, percents) of positive rational numbers without the use of a calculator.	Mission 3, Topic D, Lesson 10-16
6.RP.2	Understand the concept of a unit rate and use terms related to rate in the context of a ratio relationship.	Mission 2, Topic C, Lesson 10 Mission 3, Topic A, Lesson 1 Mission 3, Topic C, Lesson 5-7 Mission 9, Topic A, Lesson 6
6.RP.3	Make tables of equivalent ratios relating quantities with whole-number measurements, find missing values in the tables, and plot the pairs of values on the coordinate plane.	Mission 2, Topic D, Lesson 11-13
6.RP.4	Solve real-world and other mathematical problems involving rates and ratios using models and strategies such as reasoning about tables of equivalent ratios, tape diagrams, double number line diagrams, or equations.	Mission 2, Topic C, Lesson 6-10 Mission 2, Topic D, Lesson 11-14 Mission 2, Topic E, Lesson 15-16 Mission 2, Topic F, Lesson 17 Mission 3, Topic B, Lesson 3-4 Mission 3, Topic C, Lesson 5-9 Mission 3, Topic D, Lesson 10-15 Mission 6, Topic D, Lesson 16 Mission 6, Topic E, Lesson 17 Mission 9, Topic A, Lesson 4-6
6.RP.5	Use variables to represent two quantities in a proportional relationship in a real-world problem; write an equation to express one quantity, the dependent variable, in terms of the other quantity, the independent variable. Analyze the relationship between the dependent and independent variables using graphs and tables, and relate these to the equation.	Mission 6, Topic D, Lesson 16-18

INDIANA'S ACADEMIC STANDARDS FOR MATHEMATICS		ZEARN MATH
6th Grade Standards		Lessons
Algebra & Functions		
6.AF.1	Define and use multiple variables when writing expressions to represent real-world and other mathematical problems, and evaluate them for given values.	Mission 6, Topic A, Lesson 1, 3-5 Mission 6, Topic B, Lesson 6-7 Mission 7, Topic B, Lesson 8, 10
6.AF.2	Demonstrate which values from a specified set, if any, make the equation or inequality true. Use substitution to determine whether a given number in a specified set makes an equation or inequality true.	Mission 6, Topic A, Lesson 2-5 Mission 6, Topic B, Lesson 8 Mission 6, Topic C, Lesson 15 Mission 7, Topic B, Lesson 9-10
6.AF.3	Solve equations of the form $x + p = q$, $x - p = q$, $px = q$, and $x/p = q$ fluently for cases in which p , q and x are all nonnegative rational numbers. Represent real-world problems using equations of these forms and solve such problems.	Mission 6, Topic A, Lesson 3-5 Mission 6, Topic B, Lesson 7
6.AF.4	Write an inequality of the form $x > c$, $x \geq c$, $x < c$, or $x \leq c$, where c is a rational number, to represent a constraint or condition in a real-world or other mathematical problem. Explain that inequalities have infinitely many solutions and how to represent solutions on a number line diagram.	Mission 7, Topic B, Lesson 8-10
6.AF.5	Solve real-world and other mathematical problems by graphing points with rational number coordinates on a coordinate plane. Include the use of coordinates and absolute value to find distances between points with the same first coordinate or the same second coordinate.	Mission 7, Topic C, Lesson 11, 13-15
Geometry & Measurement		
6.GM.1	Convert between measurement systems (Customary to metric and metric to Customary) given the conversion factors, and use these conversions in solving real-world problems.	Mission 3, Topic B, Lesson 3-4 Mission 3, Topic C, Lesson 9
6.GM.2	Apply the sums of interior angles of triangles and quadrilaterals to solve real-world and mathematical problems.	Grade 8, Mission 1, Topic D, Lesson 15-16 <i>Zearn includes lessons on the interior angles of a triangle in</i>

INDIANA'S ACADEMIC STANDARDS FOR MATHEMATICS		ZEARN MATH
6th Grade Standards		Lessons
		<i>Grade 8, however it is recommended that teachers use supplemental materials to fully address this standard.</i>
6.GM.3	Find the area of complex shapes composed of polygons by composing or decomposing into simple shapes; apply this technique to solve real-world and other mathematical problems.	Mission 1, Topic A, Lesson 2-3 Mission 1, Topic B, Lesson 4-6 Mission 1, Topic C, Lesson 7-10 Mission 1, Topic D, Lesson 11 Mission 1, Topic G, Lesson 19 Mission 4, Topic D, Lesson 14
6.GM.4	Find the volume of a right rectangular prism with fractional edge lengths using unit cubes of the appropriate unit fraction edge lengths (e.g., using technology or concrete materials) and show that the volume is the same as would be found by multiplying the edge lengths of the prism. Apply the formulas $V = lwh$ and $V = Bh$ to find volumes of right rectangular prisms with fractional edge lengths to solve real-world and other mathematical problems.	Mission 1, Topic E, Lesson 15 Mission 4, Topic D, Lesson 14-15 Mission 4, Topic E, Lesson 17
Data Analysis & Statistics		
6.DS.1	Select, create, and interpret graphical representations of numerical data, including line plots, histograms, and box plots.	Mission 8, Topic B, Lesson 3-8 Mission 8, Topic D, Lesson 16-17
6.DS.2	Formulate statistical questions; collect and organize the data (e.g., using technology), and display and interpret the data with graphical representations (e.g., using technology).	Mission 8, Topic A, Lesson 2 Mission 8, Topic B, Lesson 3, 6-7 Mission 8, Topic D, Lesson 17
6.DS.3	Summarize numerical data sets in relation to their context in multiple ways, such as: - Report the number of observations; - Describe the nature of the attribute under investigation, including how it was measured and its units of measurement; - Determine quantitative measures of center (mean and/or median) and spread (range and interquartile range); - Describe any overall pattern and any striking deviations from the overall pattern with	Mission 8, Topic A, Lesson 2 Mission 8, Topic B, Lesson 3-7 Mission 8, Topic C, Lesson 9-12 Mission 8, Topic D, Lesson 13-17 Mission 8, Topic E, Lesson 18

INDIANA'S ACADEMIC STANDARDS FOR MATHEMATICS	ZEARN MATH
6th Grade Standards	Lessons
reference to the context in which the data were gathered; and e. Relate the choice of measures of center and spread to the shape of the data distribution and the context in which the data were gathered.	

7th Grade

INDIANA'S ACADEMIC STANDARDS FOR MATHEMATICS		ZEARN MATH
7th Grade Standards		Lessons
Number Sense		
7.NS.1	Show on a number line that a number and its opposite have a sum of 0 (are additive inverses). Find and interpret sums of rational numbers in real-world contexts.	Mission 5, Topic A, Lesson 1 Mission 5, Topic B, Lesson 2-3
7.NS.2	Show that the distance between two rational numbers on the number line is the absolute value of their difference, and apply this principle in real-world contexts.	Mission 5, Topic A, Lesson 1 Mission 5, Topic B, Lesson 2, 5-7 Mission 6, Topic D, Lesson 18
7.NS.3	Use the properties of operations, particularly the distributive property, leading to products such as $(-1)(-1) = 1$ and the rules for multiplying signed numbers.	Mission 5, Topic C, Lesson 8-9
7.NS.4	Explain that if p and q are integers, then $-(p/q) = (-p)/q = p/(-q)$ for all nonzero integers.	Mission 5, Topic C, Lesson 11
7.NS.5	Find the prime factorization of whole numbers and write the results using exponents.	<i>It is recommended that teachers use supplemental materials to fully address this standard.</i>
7.NS.6	Apply the inverse relationship between squaring and finding the square root of a perfect square whole number. Find square roots of perfect square whole numbers.	Grade 8, Mission 8, Topic B, Lesson 2, 4-5
7.NS.7	Compute fluently with rational numbers using an algorithmic approach.	Mission 4, Topic A, Lesson 5 Mission 5, Topic A, Lesson 1 Mission 5, Topic B, Lesson 2-7 Mission 5, Topic C, Lesson 8-11 Mission 6, Topic D, Lesson 18 Mission 7, Topic B, Lesson 6 Mission 8, Topic D, Lesson 16 Mission 9, Topic A, Lesson 4
Ratios & Proportional Reasoning		

INDIANA'S ACADEMIC STANDARDS FOR MATHEMATICS		ZEARN MATH
7th Grade Standards		Lessons
7.RP.1	Identify the unit rate or constant of proportionality in tables, graphs, equations, and verbal descriptions of proportional relationships.	Mission 2, Topic A, Lesson 2-3 Mission 2, Topic B, Lesson 5
7.RP.2	Use proportional relationships to solve ratio and percent problems with multiple operations (e.g., simple interest, tax, markups, markdowns, gratuities, conversions within and across measurement systems, and percent increase and decrease).	Mission 3, Topic A, Lesson 5 Mission 4, Topic B, Lesson 6-9 Mission 4, Topic C, Lesson 10-15 Mission 4, Topic D, Lesson 16 Mission 9, Topic A, Lesson 1-4 Mission 9, Topic B, Lesson 6, 8 Mission 9, Topic C, Lesson 13
7.RP.3	Represent real-world and other mathematical situations that involve proportional relationships. Write equations and draw graphs to represent these proportional relationships. Apply the definition of unit rate to $y = mx$.	Mission 2, Topic A, Lesson 2-3 Mission 2, Topic B, Lesson 4-6 Mission 2, Topic C, Lesson 7-9 Mission 2, Topic D, Lesson 10-13 Mission 2, Topic E, Lesson 14-15 Mission 3, Topic A, Lesson 3, 5, 7 Mission 4, Topic A, Lesson 3-5 Mission 9, Topic B, Lesson 5
Algebra & Functions		
7.AF.1	Apply the properties of operations (e.g., identity, inverse, commutative, associative, distributive properties) to create equivalent linear expressions, including situations that involve factoring out a common number (e.g., given $2x - 10$, create an equivalent expression $2(x - 5)$). Justify each step in the process.	Mission 6, Topic D, Lesson 18-22 Mission 9, Topic B, Lesson 7
7.AF.2	Solve real-world problems with rational numbers by using one or two operations.	Mission 3, Topic C, Lesson 11 Mission 5, Topic C, Lesson 12 Mission 5, Topic F, Lesson 17 Mission 6, Topic A, Lesson 2-6 Mission 6, Topic B, Lesson 11-12
7.AF.3	Solve equations of the form $px + q = r$ and $p(x + q) = r$ fluently, where p , q , and r are specific rational numbers. Represent real-world problems using equations of these forms and solve such problems.	Mission 5, Topic E, Lesson 15-16 Mission 6, Topic A, Lesson 4-5 Mission 6, Topic B, Lesson 10-12 Mission 7, Topic B, Lesson 7

INDIANA'S ACADEMIC STANDARDS FOR MATHEMATICS		ZEARN MATH
7th Grade Standards		Lessons
7.AF.4	Solve inequalities of the form $px + q (> \text{ or } \geq) r$ or $px + q (< \text{ or } \leq) r$, where p , q , and r are specific rational numbers. Represent real-world problems using inequalities of these forms and solve such problems. Graph the solution set of the inequality and interpret it in the context of the problem.	Mission 6, Topic C, Lesson 14, 16-27
7.AF.5	Define slope as vertical change for each unit of horizontal change, and apply that a constant rate of change or constant slope describes a linear function. Identify and describe situations with constant or varying rates of change.	Grade 8, Mission 2, Topic C, Lesson 10 Grade 8, Mission 5, Topic C, Lesson 8
7.AF.6	Graph a line given its slope and a point on the line. Find the slope of a line given its graph.	Grade 8, Mission 2, Topic C, Lesson 10
Geometry & Measurement		
7.GM.1	Solve real-world and other mathematical problems involving scale drawings of geometric figures, including computing actual lengths and areas from a scale drawing. Create a scale drawing by using proportional reasoning.	Mission 1, Topic A, Lesson 1-6 Mission 1, Topic B, Lesson 7-12 Mission 1, Topic C, Lesson 13 Mission 2, Topic A, Lesson 1 Mission 3, Topic B, Lesson 6 Mission 3, Topic C, Lesson 11 Mission 9, Topic A, Lesson 4 Mission 9, Topic C, Lesson 13
7.GM.2	Understand the formulas for area and circumference of a circle and use them to solve real-world and other mathematical problems; give an informal derivation of the relationship between circumference and area of a circle.	Mission 3, Topic A, Lesson 3-5 Mission 3, Topic B, Lesson 7-9 Mission 3, Topic C, Lesson 10-11 Mission 9, Topic A, Lesson 4 Mission 9, Topic C, Lesson 11-12
7.GM.3	Solve real-world and other mathematical problems involving volume of cylinders and three-dimensional objects composed of right rectangular prisms.	Mission 7, Topic C, Lesson 12-13, 15-16
Data Analysis, Statistics, & Probability		
7.DSP.1	Understand that statistics can be used to gain information about a population by examining a sample of the population. Understand that	Mission 8, Topic C, Lesson 12-14 Mission 8, Topic D, Lesson 15 Mission 8, Topic E, Lesson 20

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7th Grade Standards		Lessons
	conclusions and generalizations about a population from a sample are valid only if the sample is representative of that population and that random sampling tends to produce representative samples and support valid inferences.	
7.DSP.2	Find, use, and interpret measures of central tendency (mean and median) and measures of spread (range, interquartile range, and mean absolute deviation) for numerical data from random samples to draw comparative inferences about two populations.	Mission 8, Topic D, Lesson 15-16, 18-19 Mission 8, Topic E, Lesson 20 Mission 9, Topic A, Lesson 3
7.DSP.3	Make observations about the degree of visual overlap of two numerical data distributions represented in line plots or box plots. Describe how data, particularly outliers, added to a data set may affect the mean and/or median.	Mission 8, Topic C, Lesson 11 Mission 8, Topic D, Lesson 18
7.DSP.4	Understand that the probability of a chance event is a number between 0 and 1 that expresses the likelihood of the event occurring. Understand that a probability near 0 indicates an unlikely event, a probability around $\frac{1}{2}$ indicates an event that is neither unlikely nor likely, and a probability near 1 indicates a likely event. Understand that a probability of 1 indicates an event certain to occur and a probability of 0 indicates an event impossible to occur. Identify probabilities of events as impossible, unlikely, equally likely, likely, or certain.	Mission 8, Topic A, Lesson 2-6
7.DSP.5	Develop probability models that include the sample space and probabilities of outcomes to represent simple events with equally likely outcomes. Predict the approximate relative frequency of the event based on the model. Compare probabilities from the model to observed frequencies, evaluate the level of agreement, and explain possible sources of discrepancy.	Mission 8, Topic A, Lesson 3-6 Mission 8, Topic C, Lesson 14 Mission 8, Topic E, Lesson 20

8th Grade

INDIANA'S ACADEMIC STANDARDS FOR MATHEMATICS		ZEARN MATH
8th Grade Standards		Lessons
Number Sense		
8.NS.1	Give examples of rational and irrational numbers, and explain the difference between them. State decimal equivalents for any number. For rational numbers, show that the decimal equivalent terminates or repeats, and convert a repeating decimal into a rational number.	Mission 8, Topic E, Lesson 14-15
8.NS.2	Use rational approximations of irrational numbers to compare the size of irrational numbers, plot them approximately on a number line, and estimate the value of expressions involving irrational numbers.	Mission 8, Topic A, Lesson 1 Mission 8, Topic B, Lesson 4-5 Mission 8, Topic D, Lesson 12-13
8.NS.3	Given a numeric expression with common rational number bases and integer exponents, apply the properties of exponents to generate equivalent expressions.	Mission 7, Topic B, Lesson 2-8 Mission 7, Topic C, Lesson 11, 14
8.NS.4	Solve real-world problems with rational numbers by using multiple operations.	<i>It is recommended that teachers use supplemental materials to fully address this standard.</i>
Algebra & Functions		
8.AF.1	Solve linear equations and inequalities with rational number coefficients fluently, including those whose solutions require expanding expressions using the distributive property and collecting like terms. Represent real-world problems using linear equations and inequalities in one variable and solve such problems.	Mission 4, Topic B, Lesson 4, 6 Mission 7, Topic B, Lesson 5 <i>It is recommended that teachers use supplemental materials to fully address this standard.</i>
8.AF.2	Generate linear equations in one variable with one solution, infinitely many solutions, or no solutions. Justify the classification given.	Mission 4, Topic B, Lesson 3, 6-7
8.AF.3	Understand that a function assigns to each x -value (independent variable) exactly one y -value	Mission 5, Topic A, Lesson 1-2 Mission 5, Topic B, Lesson 3-5

INDIANA'S ACADEMIC STANDARDS FOR MATHEMATICS		ZEARN MATH
8th Grade Standards		Lessons
	(dependent variable), and that the graph of a function is the set of ordered pairs (x,y) .	Mission 5, Topic E, Lesson 17 Mission 9, Topic A, Lesson 1
8.AF.4	Describe qualitatively the functional relationship between two quantities by analyzing a graph (e.g., where the function is increasing or decreasing, linear or nonlinear, has a maximum or minimum value). Sketch a graph that exhibits the qualitative features of a function that has been verbally described.	Mission 5, Topic B, Lesson 5-6 Mission 5, Topic C, Lesson 10
8.AF.5	Interpret the equation $y = mx + b$ as defining a linear function whose graph is a straight line; give examples of functions that are not linear. Describe similarities and differences between linear and nonlinear functions from tables, graphs, verbal descriptions, and equations.	Mission 5, Topic B, Lesson 4, 7 Mission 5, Topic C, Lesson 8 Mission 5, Topic E, Lesson 18
8.AF.6	Construct a function to model a linear relationship between two quantities given a verbal description, table of values, or graph. Within the context of a problem, describe the meaning of m (rate of change) and b (y-intercept) in $y = mx + b$.	Mission 5, Topic C, Lesson 8-10 Mission 5, Topic D, Lesson 11
8.AF.7	Compare properties of two linear functions given in different forms, such as a table of values, equation, verbal description, and graph (e.g., compare a distance-time graph to a distance-time equation to determine which of two moving objects has greater speed).	Mission 5, Topic B, Lesson 7 Mission 5, Topic C, Lesson 8
8.AF.8	Approximate the solution of a system of equations by graphing and interpreting the reasonableness of the approximation.	Mission 3, Topic C, Lesson 13-14 Mission 4, Topic B, Lesson 9 Mission 4, Topic C, Lesson 10-15 Mission 4, Topic D, Lesson 16
Geometry & Measurement		
8.GM.1	Explore dilations, translations, rotations, and reflections on two-dimensional figures in the coordinate plane.	Mission 1, Topic A, Lesson 5-6 Mission 2, Topic A, Lesson 4-5 Mission 2, Topic C, Lesson 12

INDIANA'S ACADEMIC STANDARDS FOR MATHEMATICS		ZEARN MATH
8th Grade Standards		Lessons
8.GM.2	Solve real-world and other mathematical problems involving volume of cones, spheres, and pyramids and surface area of spheres.	Mission 5, Topic D, Lesson 13-16 Mission 5, Topic E, Lesson 17-21 Mission 5, Topic F, Lesson 22 <i>It is recommended that teachers use supplemental materials to fully address this standard.</i>
8.GM.3	Apply the Pythagorean Theorem to determine unknown side lengths in right triangles in real-world and other mathematical problems in two dimensions.	Mission 8, Topic C, Lesson 6-8, 10
Data Analysis, Statistics, & Probability		
8.DSP.1	Construct and interpret scatter plots for bivariate measurement data to investigate patterns of association between two quantitative variables. Describe patterns such as clustering, outliers, positive or negative association, linear association, and nonlinear association.	Mission 6, Topic A, Lesson 1-2 Mission 6, Topic B, Lesson 3-8
8.DSP.2	Write and use equations that model linear relationships to make predictions, including interpolation and extrapolation, in real-world situations involving bivariate measurement data. Interpret the slope and y-intercept in context.	Mission 6, Topic B, Lesson 6, 8 <i>It is recommended that teachers use supplemental materials to fully address this standard.</i>
8.DSP.3	Represent sample spaces and find probabilities of compound events (independent and dependent) using organized lists, tables, and tree diagrams.	Grade 7, Mission 8, Topic B, Lesson 7-10
8.DSP.4	Define the probability of a compound event, just as with simple events, as the fraction of outcomes in the sample space for which the compound event occurs. Use appropriate terminology to describe independent, dependent, complementary, and mutually exclusive events.	Grade 7, Mission 8, Topic B, Lesson 9
8.DSP.5	For events with a large number of outcomes, understand the use of the multiplication counting principle. Develop the multiplication counting principle, and apply it to situations with a large number of outcomes.	<i>It is recommended that teachers use supplemental materials to fully address this standard.</i>